



R Series:

mlr3

Package

In this 24th post in our ongoing exploration of the R series, we dive into the mlr3 package.

The mlr3 package offers an object-oriented programming implementation for machine learning in R. It is a successor to the mlr package, and has only the essential building blocks for machine learning. It moves away from R's S3 classes and uses the R6 encapsulated classes with reference semantics. The software contains only computational code and algorithms, and leaves the visualisation to extra packages. The return types are documented, and type safety is provided. It uses *data.table* for representing tabular data and is highly efficient and performant. The number of package dependencies is greatly reduced, and hence installation and maintenance are easier compared to the mlr package. We will use R version 4.2.2 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
```

```
R version 4.2.2 (2022-10-31) -- "Innocent and Trusting"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under the terms of the
GNU General Public License versions 2 or 3.
For more information about these matters see
https://www.gnu.org/licenses/.
```

To begin, you can install and load the 'mlr3' package using these commands:

```
> install.packages("mlr3")
```

```
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1'
```

```
(as 'lib' is unspecified)
...
*** installing help indices
*** copying figures
** building package indices
** testing if installed package can be loaded from temporary
location
** testing if installed package can be loaded from final
location
** testing if installed package keeps a record of temporary
installation path
* DONE (mlr3)

> library("mlr3")
```

mlr_tasks

The tabular data and metadata information for a machine learning problem are stored in objects called 'tasks' in mlr3. There are 11 data sets available in this version of the mlr3 package as shown here:

```
> mlr_tasks

<DictionaryTask> with 11 stored values
Keys: boston_housing, breast_cancer, german_credit, iris,
mtcars,

penguins, pima, sonar, spam, wine, zoo
```

You can obtain a task from the dictionary using the *tsk()* function. An example task for the mtcars data set is given below:

```
> task_mtcars = tsk("mtcars")
```